Rent price prediction in Berlin

Model training and picking the most suitable model

```
[2]: import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import altair as alt
from sklearn.model_selection import GridSearchCV
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.compose import TransformedTargetRegressor
from IPython.display import display, HTML
from IPython.display import Markdown as md
```

```
[3]: import sys
import os

sys.path.append(os.path.abspath("../models"))

from rent_pipeline import (
    load_data,
    split_data,
    make_cv,
    evaluate_model,
    save_bundle,
    load_bundle,
    predict_listing,
    build_preprocessor,
    build_pipeline
)
```

```
[4]: df = load_data('../datasets/rent_listings_final.csv')
df.shape
```

```
[4]: (15588, 23)
```

```
[5]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15588 entries, 0 to 15587
Data columns (total 23 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   listing_url          15588 non-null  object
 1   cold_rent            15588 non-null  float64
 2   area                 15588 non-null  float64
 3   rooms                15588 non-null  float64
 4   floor                13646 non-null  float64
 5   free_from            15588 non-null  object
 6   has_balkon           15588 non-null  bool
 7   has_terrasse         15588 non-null  bool
 8   has_garten           15588 non-null  bool
 9   elevator             15588 non-null  bool
 10  parking              15588 non-null  bool
 11  has_basement         15588 non-null  bool
 12  is_barrier-free      15588 non-null  bool
 13  has_built-in_kitchen 15588 non-null  bool
 14  has_bathtub          15588 non-null  bool
 15  has_shower           15588 non-null  bool
 16  flooring_type        15588 non-null  object
 17  energy_source        15588 non-null  object
 18  heating_type         15588 non-null  object
 19  property_condition   15588 non-null  object
 20  year_built           7662 non-null   float64
 21  zip_code             15588 non-null  int64
 22  district             15588 non-null  object
dtypes: bool(10), float64(5), int64(1), object(7)
memory usage: 1.7+ MB
```

```
[6]: df = df.drop(columns=["listing_url","zip_code"])
```

```
[7]: # So we get reproducable results:
np.random.seed(16)
```

Part 1: Modeling and model evaluation

1. Split the dataset into a train set (80%) and a test set (20%).

```
[8]: X_train, X_test, y_train, y_test = split_data(df, test_size=0.2, random_state=16)
```

2. Modeling and training

```
[15]: from sklearn.linear_model import LinearRegression

metrics_linear, bestmodel_linear = evaluate_model(
    "linear_regression",
    LinearRegression(),
    {},
    X_train,
    y_train,
    X_test,
    y_test
)

metrics_linear
```

```
[15]: {'model': 'linear_regression',
 'best_params': {},
 'cv_rmse': 432.648392221836,
 'train_rmse': 428.58787681245644,
 'test_rmse': 408.98770317538276,
 'train_r2': 0.6663174481981542,
 'test_r2': 0.6796527825087897,
 'test_mae': 289.3131410640148,
 'overfit_gap_r2': -0.013335334310635494}
```

```
[16]: display(md(f'<span style="font-weight:500">Linear regression achieved an R² score of {metrics_linear["test_r2"]:.4f}, which is not very good. However, this
```

Linear regression achieved an R² score of 0.6797, which is not very good. However, this is expected because a simple linear model cannot capture the complex nonlinear relationships present in rental prices.

```
[17]: from sklearn.neighbors import KNeighborsRegressor

metrics_knn, bestmodel_knn = evaluate_model(
    "knn",
    KNeighborsRegressor(algorithm="ball_tree"),
    {
        "model__n_neighbors": [3, 5],
        "model__weights": ["distance"],
    },
    X_train,
    y_train,
    X_test,
    y_test
)

metrics_knn
```

```
[17]: {'model': 'knn',
 'best_params': {'model__n_neighbors': 3, 'model__weights': 'distance'},
 'cv_rmse': 295.7847761100729,
 'train_rmse': 46.105525357852365,
 'test_rmse': 265.5247932558738,
 'train_r2': 0.9961384675957481,
 'test_r2': 0.8649761105915468,
 'test_mae': 153.74112245098948,
 'overfit_gap_r2': 0.1311623570042013}
```

```
[18]: display(md(f'<span style="font-weight:500">KNN performs slightly better, with an R² score of {metrics_knn["test_r2"]:.4f} , but not good enough to be consid
```

KNN performs slightly better, with an R² score of 0.8650 , but not good enough to be considered our final model.

```
[19]: from sklearn.svm import SVR

metrics_svr, bestmodel_svr = evaluate_model(
    "SVR",
    TransformedTargetRegressor(
        regressor=SVR(),
        transformer=StandardScaler()
    ),
    {
        "model__regressor__kernel": ["linear"],
        "model__regressor__C": [0.1, 1, 10, 100],
        "model__regressor__epsilon": [0.01, 0.05, 0.1, 0.2, 0.5]
    },
    X_train,
    y_train,
    X_test,
    y_test
)

metrics_svr
```

```
[19]: {'model': 'SVR',
 'best_params': {'model__regressor__C': 0.1,
  'model__regressor__epsilon': 0.5,
  'model__regressor__kernel': 'linear'},
 'cv_rmse': 437.011182138511,
 'train_rmse': 434.31287558571347,
 'test_rmse': 408.3316662743395,
 'train_r2': 0.657343367322307,
 'test_r2': 0.6806796664462924,
 'test_mae': 286.55976382177033,
 'overfit_gap_r2': -0.023336297123985328}
```

SVR performs very poorly in this dataset

```
[20]: from sklearn.ensemble import RandomForestRegressor

metrics_rf, bestmodel_rf = evaluate_model(
    "Random Forest",
    RandomForestRegressor(random_state=42),
    {
        "model__n_estimators": [100, 200, 300, 400],
        "model__max_depth": [None, 10, 20, 30, 40],
        "model__min_samples_split": [2, 5, 10, 20],
        "model__min_samples_leaf": [1, 2, 4, 8],
        "model__max_features": ["sqrt", "log2"],
    },
    X_train,
    y_train,
    X_test,
    y_test
)

metrics_rf
```

```
[20]: {'model': 'Random Forest',
 'best_params': {'model__max_depth': 30,
  'model__max_features': 'sqrt',
  'model__min_samples_leaf': 1,
  'model__min_samples_split': 2,
  'model__n_estimators': 200},
 'cv_rmse': 234.58062413897596,
 'train_rmse': 108.98730031958362,
 'test_rmse': 209.19388040533767,
 'train_r2': 0.9784222814781925,
 'test_r2': 0.916189504775013,
 'test_mae': 125.33442477938395,
 'overfit_gap_r2': 0.062232776703179504}
```

```
[21]: from IPython.display import Markdown as md
display(md(f'<span style="font-weight:500">We got a good R squared of {metrics_rf["test_r2"]:.4f} and lower errors. This is the best model so far. We shoul
```

We got a good R squared of 0.9162 and lower errors. This is the best model so far. We should aim for a higher R-squared.

```
[23]: from xgboost import XGBRegressor

param_grid_xgb = {
    "model__n_estimators": [400, 600],
    "model__max_depth": [5, 7],
    "model__learning_rate": [0.001, 0.01, 0.1],
    "model__subsample": [0.8, 1.0],
    "model__colsample_bytree": [0.8, 1.0],
    "model__min_child_weight": [1, 3],
    "model__gamma": [0, 0.1]
}

metrics_xgb, bestmodel_xgb = evaluate_model(
    "XGBoost",
    XGBRegressor(
        random_state=42,
        objective="reg:squarederror"
    ),
    param_grid_xgb,
    X_train,
    y_train,
    X_test,
    y_test
)

metrics_xgb
```

```
[23]: {'model': 'XGBoost',
 'best_params': {'model__colsample_bytree': 0.8,
  'model__gamma': 0.1,
  'model__learning_rate': 0.1,
  'model__max_depth': 7,
  'model__min_child_weight': 1,
  'model__n_estimators': 600,
  'model__subsample': 1.0},
 'cv_rmse': 213.5257409699028,
 'train_rmse': 115.66084088666365,
 'test_rmse': 184.13917281244596,
 'train_r2': 0.9756988724814845,
 'test_r2': 0.9350629087994863,
 'test_mae': 111.74026820272724,
 'overfit_gap_r2': 0.04063596368199818}
```

Both XGBoost and RandomForest have a really good performance with XGBoost having a slighlty higher R-squared of almost 0.935, which, considering our data set, is a very good and realistic performance.
Personally, I thought that XGBoost was going to be the best expected model and most likely this is the model we are going to deploy.
If you can notice, the hyperparameters of the best-performing model are the highest values, so after picking the final model, I would prefer to try it with other values one last time and picking the ones that don't cause overfitting.

```
[24]: import tensorflow as tf
from tensorflow.keras import layers, models, callbacks

def build_nn(input_dim):

    model = models.Sequential()

    model.add(layers.Input(shape=(input_dim,)))

    model.add(layers.Dense(256, activation='relu',
              kernel_regularizer=tf.keras.regularizers.l2(0.001)))
    model.add(layers.BatchNormalization())
    model.add(layers.Dropout(0.2))

    model.add(layers.Dense(128, activation='relu',
              kernel_regularizer=tf.keras.regularizers.l2(0.001)))
    model.add(layers.BatchNormalization())
    model.add(layers.Dropout(0.2))

    model.add(layers.Dense(64, activation='relu',
              kernel_regularizer=tf.keras.regularizers.l2(0.001)))

    model.add(layers.Dense(32, activation='relu',
              kernel_regularizer=tf.keras.regularizers.l2(0.001)))

    model.add(layers.Dense(1))

    model.compile(
        optimizer=tf.keras.optimizers.Adam(learning_rate = 0.0005),
        loss='mse',
        metrics=[tf.keras.metrics.RootMeanSquaredError()]
    )

    return model
```

```
[29]: from sklearn.metrics import (
    mean_squared_error,
    mean_absolute_error,
    r2_score,
)


def evaluate_nn(
    X_train,
    y_train,
    X_test,
    y_test,
    epochs=300,
    batch_size=64,
    feature_cols=None
):
    nn_pipeline = Pipeline([
        ("preprocessor", build_preprocessor(feature_cols))
    ])

    X_train_processed = nn_pipeline.fit_transform(X_train)
    X_test_processed = nn_pipeline.transform(X_test)

    nn_model = build_nn(X_train_processed.shape[1])

    early_stop = callbacks.EarlyStopping(
        monitor="val_loss",
        patience=20,
        restore_best_weights=True
    )

    history = nn_model.fit(
        X_train_processed,
        y_train,
        validation_split=0.2,
        epochs=epochs,
        batch_size=batch_size,
        callbacks=[early_stop],
        verbose=1
    )

    y_pred_train = nn_model.predict(
        X_train_processed, verbose=0
    ).ravel()

    y_pred_test = nn_model.predict(
        X_test_processed, verbose=0
    ).ravel()

    train_r2 = r2_score(y_train, y_pred_train)
    test_r2 = r2_score(y_test, y_pred_test)

    metrics = {
        "model": "Neural Network",
        "best_params": {
            "epochs": epochs,
            "batch_size": batch_size
        },
        "cv_rmse": None,
        "train_rmse": float(
            np.sqrt(mean_squared_error(y_train, y_pred_train))
        ),
        "test_rmse": float(
            np.sqrt(mean_squared_error(y_test, y_pred_test))
        ),
        "train_r2": float(train_r2),
        "test_r2": float(test_r2),
        "test_mae": float(
            mean_absolute_error(y_test, y_pred_test)
        ),
        "overfit_gap_r2": float(train_r2 - test_r2),
    }

    return metrics, nn_model, nn_pipeline, history
```

```
[30]: metrics_nn, bestmodel_nn, nn_pipeline, history_nn = evaluate_nn(
    X_train,
    y_train,
    X_test,
    y_test,
    epochs=300,
    batch_size=64
)

metrics_nn
```

```
Epoch 1/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 9s 13ms/step - loss: 1934628.7500 - root_mean_squared_error: 1390.9092 - val_loss: 1721823.0000 - val_root_mean_squared_error:
1312.1825
Epoch 2/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 8ms/step - loss: 685842.2500 - root_mean_squared_error: 828.1556 - val_loss: 323690.6562 - val_root_mean_squared_error: 56
8.9376
Epoch 3/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 10ms/step - loss: 257488.2188 - root_mean_squared_error: 507.4323 - val_loss: 195937.2188 - val_root_mean_squared_error: 44
2.6471
Epoch 4/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 10ms/step - loss: 188379.7812 - root_mean_squared_error: 434.0264 - val_loss: 152753.9062 - val_root_mean_squared_error: 39
0.8364
Epoch 5/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 11ms/step - loss: 165258.9375 - root_mean_squared_error: 406.5195 - val_loss: 124662.1562 - val_root_mean_squared_error: 35
3.0741
Epoch 6/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 137505.7656 - root_mean_squared_error: 370.8165 - val_loss: 105146.4219 - val_root_mean_squared_error: 32
4.2616
Epoch 7/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 123827.5312 - root_mean_squared_error: 351.8901 - val_loss: 98293.2891 - val_root_mean_squared_error: 313.
5162
Epoch 8/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 112957.6484 - root_mean_squared_error: 336.0903 - val_loss: 92738.8281 - val_root_mean_squared_error: 304.
5291
Epoch 9/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 108685.8672 - root_mean_squared_error: 329.6740 - val_loss: 88567.6953 - val_root_mean_squared_error: 297.
6017
Epoch 10/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 105590.8594 - root_mean_squared_error: 324.9460 - val_loss: 85379.0625 - val_root_mean_squared_error: 292.
1953
Epoch 11/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 12ms/step - loss: 98064.3047 - root_mean_squared_error: 313.1507 - val_loss: 83209.8125 - val_root_mean_squared_error: 288.
4594
Epoch 12/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 3s 18ms/step - loss: 91389.9922 - root_mean_squared_error: 302.3062 - val_loss: 80790.9531 - val_root_mean_squared_error: 284.
2350
Epoch 13/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 3s 16ms/step - loss: 91742.6875 - root_mean_squared_error: 302.8890 - val_loss: 81155.5312 - val_root_mean_squared_error: 284.
8764
Epoch 14/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 12ms/step - loss: 90994.7031 - root_mean_squared_error: 301.6516 - val_loss: 76454.5078 - val_root_mean_squared_error: 276.
5023
Epoch 15/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 84922.7422 - root_mean_squared_error: 291.4134 - val_loss: 76777.6562 - val_root_mean_squared_error: 277.0
860
Epoch 16/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 81068.0703 - root_mean_squared_error: 284.7228 - val_loss: 78508.6641 - val_root_mean_squared_error: 280.1
922
Epoch 17/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 82690.9922 - root_mean_squared_error: 287.5587 - val_loss: 75881.5859 - val_root_mean_squared_error: 275.4
643
Epoch 18/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 79579.9453 - root_mean_squared_error: 282.0974 - val_loss: 75731.6953 - val_root_mean_squared_error: 275.1
921
Epoch 19/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 76580.5469 - root_mean_squared_error: 276.7300 - val_loss: 72829.1484 - val_root_mean_squared_error: 269.8
668
Epoch 20/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 11ms/step - loss: 75498.5859 - root_mean_squared_error: 274.7682 - val_loss: 74022.2266 - val_root_mean_squared_error: 272.
0684
Epoch 21/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 11ms/step - loss: 73254.5938 - root_mean_squared_error: 270.6539 - val_loss: 73114.3750 - val_root_mean_squared_error: 270.
3948
Epoch 22/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 9ms/step - loss: 74348.7578 - root_mean_squared_error: 272.6677 - val_loss: 73809.5703 - val_root_mean_squared_error: 271.6
772
Epoch 23/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 8ms/step - loss: 71522.5312 - root_mean_squared_error: 267.4350 - val_loss: 72096.0547 - val_root_mean_squared_error: 268.5
051
Epoch 24/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 10ms/step - loss: 69090.1562 - root_mean_squared_error: 262.8481 - val_loss: 70172.7969 - val_root_mean_squared_error: 264.
8995
Epoch 25/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 11ms/step - loss: 70045.8672 - root_mean_squared_error: 264.6598 - val_loss: 73161.4609 - val_root_mean_squared_error: 270.
4818
Epoch 26/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 11ms/step - loss: 69931.6016 - root_mean_squared_error: 264.4438 - val_loss: 69684.8359 - val_root_mean_squared_error: 263.
9768
Epoch 27/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 11ms/step - loss: 70320.6016 - root_mean_squared_error: 265.1783 - val_loss: 70151.5234 - val_root_mean_squared_error: 264.
8593
Epoch 28/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 13ms/step - loss: 69803.6094 - root_mean_squared_error: 264.2016 - val_loss: 69221.9609 - val_root_mean_squared_error: 263.
0986
Epoch 29/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 3s 18ms/step - loss: 65534.1016 - root_mean_squared_error: 255.9942 - val_loss: 68261.2109 - val_root_mean_squared_error: 261.
2664
Epoch 30/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 3s 16ms/step - loss: 65032.5117 - root_mean_squared_error: 255.0125 - val_loss: 70389.0781 - val_root_mean_squared_error: 265.
3073
Epoch 31/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 13ms/step - loss: 66305.8359 - root_mean_squared_error: 257.4970 - val_loss: 68414.4375 - val_root_mean_squared_error: 261.
5594
Epoch 32/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 11ms/step - loss: 65315.9336 - root_mean_squared_error: 255.5676 - val_loss: 69677.1484 - val_root_mean_squared_error: 263.
9622
Epoch 33/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 64452.8750 - root_mean_squared_error: 253.8735 - val_loss: 66765.5781 - val_root_mean_squared_error: 258.3
882
Epoch 34/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 9ms/step - loss: 63543.0391 - root_mean_squared_error: 252.0752 - val_loss: 68795.9375 - val_root_mean_squared_error: 262.2
877
Epoch 35/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 63799.7812 - root_mean_squared_error: 252.5840 - val_loss: 69530.3203 - val_root_mean_squared_error: 263.6
839
Epoch 36/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 61494.5312 - root_mean_squared_error: 247.9786 - val_loss: 65641.2344 - val_root_mean_squared_error: 256.2
032
Epoch 37/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 13ms/step - loss: 63426.7969 - root_mean_squared_error: 251.8445 - val_loss: 66397.7031 - val_root_mean_squared_error: 257.
6753
Epoch 38/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 10ms/step - loss: 61581.1523 - root_mean_squared_error: 248.1532 - val_loss: 66698.6719 - val_root_mean_squared_error: 258.
2586
Epoch 39/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 11ms/step - loss: 61186.6367 - root_mean_squared_error: 247.3570 - val_loss: 65764.2578 - val_root_mean_squared_error: 256.
4432
Epoch 40/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 59186.7969 - root_mean_squared_error: 243.1165 - val_loss: 67602.2812 - val_root_mean_squared_error: 260.0
021
Epoch 41/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 60268.2773 - root_mean_squared_error: 245.4936 - val_loss: 70634.9688 - val_root_mean_squared_error: 265.7
702
Epoch 42/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 58054.0352 - root_mean_squared_error: 240.9416 - val_loss: 65503.0781 - val_root_mean_squared_error: 255.9
334
Epoch 43/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 58756.7031 - root_mean_squared_error: 242.3954 - val_loss: 66444.5156 - val_root_mean_squared_error: 257.7
660
Epoch 44/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 8ms/step - loss: 59348.6016 - root_mean_squared_error: 243.6133 - val_loss: 66248.4062 - val_root_mean_squared_error: 257.3
853
Epoch 45/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 60502.4922 - root_mean_squared_error: 245.9701 - val_loss: 65400.8906 - val_root_mean_squared_error: 255.7
336
Epoch 46/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 59136.3672 - root_mean_squared_error: 243.1772 - val_loss: 64732.6172 - val_root_mean_squared_error: 254.4
237
Epoch 47/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 14ms/step - loss: 59026.1797 - root_mean_squared_error: 242.9506 - val_loss: 65669.5078 - val_root_mean_squared_error: 256.
2582
Epoch 48/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 3s 20ms/step - loss: 56108.7539 - root_mean_squared_error: 236.8703 - val_loss: 66919.5391 - val_root_mean_squared_error: 258.
6858
Epoch 49/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 3s 16ms/step - loss: 57565.7109 - root_mean_squared_error: 239.9260 - val_loss: 65808.3281 - val_root_mean_squared_error: 256.
5289
Epoch 50/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 10ms/step - loss: 57290.9961 - root_mean_squared_error: 239.3529 - val_loss: 66952.1172 - val_root_mean_squared_error: 258.
7487
Epoch 51/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 57307.5977 - root_mean_squared_error: 239.3875 - val_loss: 64597.4805 - val_root_mean_squared_error: 254.1
579
Epoch 52/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 53545.5508 - root_mean_squared_error: 231.3964 - val_loss: 64987.3242 - val_root_mean_squared_error: 254.9
237
Epoch 53/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 55670.1094 - root_mean_squared_error: 235.9425 - val_loss: 63464.2891 - val_root_mean_squared_error: 251.9
187
Epoch 54/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 14ms/step - loss: 55366.5703 - root_mean_squared_error: 235.2984 - val_loss: 65443.6680 - val_root_mean_squared_error: 255.
8172
Epoch 55/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 9ms/step - loss: 55365.5312 - root_mean_squared_error: 235.2962 - val_loss: 63725.2539 - val_root_mean_squared_error: 252.4
361
Epoch 56/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 10ms/step - loss: 53413.7500 - root_mean_squared_error: 231.1115 - val_loss: 62574.4023 - val_root_mean_squared_error: 250.
1462
Epoch 57/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 55338.2344 - root_mean_squared_error: 235.2382 - val_loss: 64498.3867 - val_root_mean_squared_error: 253.9
629
Epoch 58/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 9ms/step - loss: 53328.6562 - root_mean_squared_error: 230.9273 - val_loss: 63436.6562 - val_root_mean_squared_error: 251.8
638
Epoch 59/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 54336.9375 - root_mean_squared_error: 233.1001 - val_loss: 63053.0625 - val_root_mean_squared_error: 251.1
012
Epoch 60/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 12ms/step - loss: 55603.6133 - root_mean_squared_error: 235.8015 - val_loss: 65375.2812 - val_root_mean_squared_error: 255.
6834
Epoch 61/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 11ms/step - loss: 53034.1367 - root_mean_squared_error: 230.2887 - val_loss: 64500.7734 - val_root_mean_squared_error: 253.
9675
Epoch 62/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 9ms/step - loss: 52972.6719 - root_mean_squared_error: 230.1552 - val_loss: 65824.6797 - val_root_mean_squared_error: 256.5
607
Epoch 63/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 14ms/step - loss: 53555.4844 - root_mean_squared_error: 231.4178 - val_loss: 62451.0547 - val_root_mean_squared_error: 249.
8995
Epoch 64/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 13ms/step - loss: 52986.5391 - root_mean_squared_error: 230.1852 - val_loss: 66204.6406 - val_root_mean_squared_error: 257.
3001
Epoch 65/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 3s 16ms/step - loss: 53266.5078 - root_mean_squared_error: 230.7926 - val_loss: 64799.9922 - val_root_mean_squared_error: 254.
5559
Epoch 66/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 14ms/step - loss: 53424.3320 - root_mean_squared_error: 231.1342 - val_loss: 65628.1797 - val_root_mean_squared_error: 256.
1774
Epoch 67/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 10ms/step - loss: 52284.1211 - root_mean_squared_error: 228.6544 - val_loss: 65049.7852 - val_root_mean_squared_error: 255.
0460
Epoch 68/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 52566.1445 - root_mean_squared_error: 229.2702 - val_loss: 66346.3359 - val_root_mean_squared_error: 257.5
753
Epoch 69/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 53413.2812 - root_mean_squared_error: 231.1103 - val_loss: 64618.0000 - val_root_mean_squared_error: 254.1
981
Epoch 70/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 52104.8789 - root_mean_squared_error: 228.2620 - val_loss: 63021.6016 - val_root_mean_squared_error: 251.0
384
Epoch 71/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 11ms/step - loss: 52301.6211 - root_mean_squared_error: 228.6926 - val_loss: 63641.8750 - val_root_mean_squared_error: 252.
2708
Epoch 72/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 10ms/step - loss: 51128.7773 - root_mean_squared_error: 226.1138 - val_loss: 61926.8711 - val_root_mean_squared_error: 248.
8484
Epoch 73/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 13ms/step - loss: 51704.9062 - root_mean_squared_error: 227.3842 - val_loss: 63815.8984 - val_root_mean_squared_error: 252.
6155
Epoch 74/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 11ms/step - loss: 52562.0078 - root_mean_squared_error: 229.2612 - val_loss: 63788.6641 - val_root_mean_squared_error: 252.
5616
Epoch 75/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 50902.5078 - root_mean_squared_error: 225.6129 - val_loss: 62133.0352 - val_root_mean_squared_error: 249.2
623
Epoch 76/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 50288.4336 - root_mean_squared_error: 224.2478 - val_loss: 64978.4844 - val_root_mean_squared_error: 254.9
061
Epoch 77/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 50838.5859 - root_mean_squared_error: 225.4712 - val_loss: 63964.1133 - val_root_mean_squared_error: 252.9
086
Epoch 78/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 49237.2578 - root_mean_squared_error: 221.8917 - val_loss: 63189.5938 - val_root_mean_squared_error: 251.3
728
Epoch 79/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 50720.3828 - root_mean_squared_error: 225.2088 - val_loss: 67574.5078 - val_root_mean_squared_error: 259.9
484
Epoch 80/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 9ms/step - loss: 49302.1602 - root_mean_squared_error: 222.0379 - val_loss: 63085.5117 - val_root_mean_squared_error: 251.1
656
Epoch 81/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 10ms/step - loss: 49761.6523 - root_mean_squared_error: 223.0701 - val_loss: 61906.9141 - val_root_mean_squared_error: 248.
8003
Epoch 82/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 12ms/step - loss: 49495.0195 - root_mean_squared_error: 222.4717 - val_loss: 61772.0625 - val_root_mean_squared_error: 248.
5371
Epoch 83/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 3s 15ms/step - loss: 51030.3281 - root_mean_squared_error: 225.8958 - val_loss: 62288.8789 - val_root_mean_squared_error: 249.
5747
Epoch 84/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 3s 17ms/step - loss: 49444.3516 - root_mean_squared_error: 222.3578 - val_loss: 63643.9766 - val_root_mean_squared_error: 252.
2749
Epoch 85/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 11ms/step - loss: 48001.5352 - root_mean_squared_error: 219.0094 - val_loss: 62273.7383 - val_root_mean_squared_error: 249.
5443
Epoch 86/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 10ms/step - loss: 49047.1992 - root_mean_squared_error: 221.4629 - val_loss: 63078.2852 - val_root_mean_squared_error: 251.
1512
Epoch 87/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 3s 13ms/step - loss: 48959.1289 - root_mean_squared_error: 221.2639 - val_loss: 66354.2969 - val_root_mean_squared_error: 257.
5906
Epoch 88/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 9ms/step - loss: 49353.6211 - root_mean_squared_error: 222.1536 - val_loss: 62963.3828 - val_root_mean_squared_error: 250.9
223
Epoch 89/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 10ms/step - loss: 49220.5352 - root_mean_squared_error: 221.8539 - val_loss: 64549.8203 - val_root_mean_squared_error: 254.
0638
Epoch 90/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 10ms/step - loss: 48194.2539 - root_mean_squared_error: 219.5287 - val_loss: 65771.8672 - val_root_mean_squared_error: 256.
4575
Epoch 91/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 48558.3398 - root_mean_squared_error: 220.3564 - val_loss: 63439.8672 - val_root_mean_squared_error: 251.8
699
Epoch 92/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 49485.5859 - root_mean_squared_error: 222.4503 - val_loss: 66199.9141 - val_root_mean_squared_error: 257.2
907
Epoch 93/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 10ms/step - loss: 47022.2930 - root_mean_squared_error: 216.8430 - val_loss: 62916.7422 - val_root_mean_squared_error: 250.
8293
Epoch 94/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 10ms/step - loss: 47054.4258 - root_mean_squared_error: 216.9171 - val_loss: 64175.3438 - val_root_mean_squared_error: 253.
3257
Epoch 95/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 11ms/step - loss: 48513.4805 - root_mean_squared_error: 220.2545 - val_loss: 64469.4258 - val_root_mean_squared_error: 253.
9055
Epoch 96/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 12ms/step - loss: 48123.7305 - root_mean_squared_error: 219.3600 - val_loss: 63255.9805 - val_root_mean_squared_error: 251.
5046
Epoch 97/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 12ms/step - loss: 47968.3438 - root_mean_squared_error: 219.0035 - val_loss: 62103.8125 - val_root_mean_squared_error: 249.
2035
Epoch 98/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 9ms/step - loss: 47732.5664 - root_mean_squared_error: 218.4746 - val_loss: 61595.2266 - val_root_mean_squared_error: 248.1
810
Epoch 99/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 3s 16ms/step - loss: 47462.3047 - root_mean_squared_error: 217.8551 - val_loss: 60494.5898 - val_root_mean_squared_error: 245.
9536
Epoch 100/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 3s 17ms/step - loss: 46936.1445 - root_mean_squared_error: 216.6442 - val_loss: 61977.0273 - val_root_mean_squared_error: 248.
9490
Epoch 101/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 14ms/step - loss: 46311.1719 - root_mean_squared_error: 215.1970 - val_loss: 62071.9609 - val_root_mean_squared_error: 249.
1396
Epoch 102/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 11ms/step - loss: 48142.3398 - root_mean_squared_error: 219.4104 - val_loss: 62010.8945 - val_root_mean_squared_error: 249.
0170
Epoch 103/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 47548.3359 - root_mean_squared_error: 218.0525 - val_loss: 61781.2852 - val_root_mean_squared_error: 248.5
555
Epoch 104/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 12ms/step - loss: 46846.2070 - root_mean_squared_error: 216.4365 - val_loss: 62259.2852 - val_root_mean_squared_error: 249.
5152
Epoch 105/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 46868.6562 - root_mean_squared_error: 216.4883 - val_loss: 60881.5508 - val_root_mean_squared_error: 246.7
389
Epoch 106/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 11ms/step - loss: 47715.1133 - root_mean_squared_error: 218.4346 - val_loss: 61723.6562 - val_root_mean_squared_error: 248.
4395
Epoch 107/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 46280.0742 - root_mean_squared_error: 215.1246 - val_loss: 60209.0586 - val_root_mean_squared_error: 245.3
723
Epoch 108/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 10ms/step - loss: 47843.2227 - root_mean_squared_error: 218.7275 - val_loss: 60512.6172 - val_root_mean_squared_error: 245.
9901
Epoch 109/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 3s 10ms/step - loss: 46943.0430 - root_mean_squared_error: 216.6600 - val_loss: 61102.9727 - val_root_mean_squared_error: 247.
1872
Epoch 110/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 9ms/step - loss: 45308.8203 - root_mean_squared_error: 212.8552 - val_loss: 66023.2188 - val_root_mean_squared_error: 256.9
470
Epoch 111/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 46090.6836 - root_mean_squared_error: 214.6840 - val_loss: 63452.0977 - val_root_mean_squared_error: 251.8
941
Epoch 112/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 44859.0234 - root_mean_squared_error: 211.7960 - val_loss: 62787.3828 - val_root_mean_squared_error: 250.5
711
Epoch 113/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 45609.4258 - root_mean_squared_error: 213.5601 - val_loss: 63372.7461 - val_root_mean_squared_error: 251.7
365
Epoch 114/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 44314.8789 - root_mean_squared_error: 210.5074 - val_loss: 61390.9219 - val_root_mean_squared_error: 247.7
689
Epoch 115/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 11ms/step - loss: 45776.2969 - root_mean_squared_error: 213.9505 - val_loss: 64142.6016 - val_root_mean_squared_error: 253.
2609
Epoch 116/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 3s 17ms/step - loss: 46071.1367 - root_mean_squared_error: 214.6384 - val_loss: 62215.2500 - val_root_mean_squared_error: 249.
4268
Epoch 117/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 3s 16ms/step - loss: 43729.9062 - root_mean_squared_error: 209.1134 - val_loss: 61364.3359 - val_root_mean_squared_error: 247.
7152
Epoch 118/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 3s 17ms/step - loss: 44949.4219 - root_mean_squared_error: 212.0092 - val_loss: 63392.8906 - val_root_mean_squared_error: 251.
7764
Epoch 119/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 44862.5703 - root_mean_squared_error: 211.8043 - val_loss: 66014.6562 - val_root_mean_squared_error: 256.9
302
Epoch 120/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 11ms/step - loss: 45580.3672 - root_mean_squared_error: 213.4920 - val_loss: 61592.2969 - val_root_mean_squared_error: 248.
1749
Epoch 121/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 10ms/step - loss: 43544.0781 - root_mean_squared_error: 208.6685 - val_loss: 60159.9180 - val_root_mean_squared_error: 245.
2720
Epoch 122/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 10ms/step - loss: 46486.4141 - root_mean_squared_error: 215.6035 - val_loss: 63429.9727 - val_root_mean_squared_error: 251.
8500
Epoch 123/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 9ms/step - loss: 45390.2656 - root_mean_squared_error: 213.0463 - val_loss: 63433.4766 - val_root_mean_squared_error: 251.8
570
Epoch 124/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 11ms/step - loss: 43247.9922 - root_mean_squared_error: 207.9578 - val_loss: 60485.9844 - val_root_mean_squared_error: 245.
9359
Epoch 125/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 44680.4219 - root_mean_squared_error: 211.3738 - val_loss: 65490.8867 - val_root_mean_squared_error: 255.9
009
Epoch 126/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 11ms/step - loss: 44412.7773 - root_mean_squared_error: 210.7398 - val_loss: 60454.0898 - val_root_mean_squared_error: 245.
8710
Epoch 127/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 44281.4570 - root_mean_squared_error: 210.4279 - val_loss: 59575.7305 - val_root_mean_squared_error: 244.0
782
Epoch 128/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 3s 13ms/step - loss: 42503.6875 - root_mean_squared_error: 206.1604 - val_loss: 57839.5312 - val_root_mean_squared_error: 240.
4953
Epoch 129/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 43893.6992 - root_mean_squared_error: 209.5045 - val_loss: 61480.7461 - val_root_mean_squared_error: 247.9
500
Epoch 130/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 10ms/step - loss: 44211.5156 - root_mean_squared_error: 210.2617 - val_loss: 61974.4414 - val_root_mean_squared_error: 248.
9435
Epoch 131/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 10ms/step - loss: 44932.4727 - root_mean_squared_error: 211.9691 - val_loss: 60802.4492 - val_root_mean_squared_error: 246.
5783
Epoch 132/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 10ms/step - loss: 42393.9766 - root_mean_squared_error: 205.8941 - val_loss: 61844.8320 - val_root_mean_squared_error: 248.
6830
Epoch 133/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 3s 16ms/step - loss: 42735.1680 - root_mean_squared_error: 206.7211 - val_loss: 59757.3438 - val_root_mean_squared_error: 244.
4500
Epoch 134/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 3s 16ms/step - loss: 43043.6680 - root_mean_squared_error: 207.4659 - val_loss: 59179.8672 - val_root_mean_squared_error: 243.
2659
Epoch 135/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 12ms/step - loss: 43573.0625 - root_mean_squared_error: 208.7378 - val_loss: 60803.0664 - val_root_mean_squared_error: 246.
5796
Epoch 136/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 11ms/step - loss: 43470.5039 - root_mean_squared_error: 208.4920 - val_loss: 59261.5703 - val_root_mean_squared_error: 243.
4337
Epoch 137/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 43145.5586 - root_mean_squared_error: 207.7112 - val_loss: 61196.3164 - val_root_mean_squared_error: 247.3
757
Epoch 138/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 12ms/step - loss: 43551.3828 - root_mean_squared_error: 208.6859 - val_loss: 63655.6602 - val_root_mean_squared_error: 252.
2975
Epoch 139/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 44029.4062 - root_mean_squared_error: 209.8281 - val_loss: 72227.1172 - val_root_mean_squared_error: 268.7
481
Epoch 140/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 10ms/step - loss: 43480.0234 - root_mean_squared_error: 208.5148 - val_loss: 62426.1250 - val_root_mean_squared_error: 249.
8490
Epoch 141/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 43374.9492 - root_mean_squared_error: 208.2627 - val_loss: 60863.2383 - val_root_mean_squared_error: 246.7
015
Epoch 142/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 43210.3164 - root_mean_squared_error: 207.8670 - val_loss: 60069.6953 - val_root_mean_squared_error: 245.0
879
Epoch 143/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 41003.5156 - root_mean_squared_error: 202.4893 - val_loss: 60685.3750 - val_root_mean_squared_error: 246.3
407
Epoch 144/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 9ms/step - loss: 41972.0820 - root_mean_squared_error: 204.8670 - val_loss: 60505.0273 - val_root_mean_squared_error: 245.9
744
Epoch 145/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 9ms/step - loss: 43739.0977 - root_mean_squared_error: 209.1351 - val_loss: 61009.5000 - val_root_mean_squared_error: 246.9
978
Epoch 146/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 12ms/step - loss: 42212.1562 - root_mean_squared_error: 205.4520 - val_loss: 64758.4023 - val_root_mean_squared_error: 254.
4736
Epoch 147/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 11ms/step - loss: 42514.6914 - root_mean_squared_error: 206.1870 - val_loss: 63230.6406 - val_root_mean_squared_error: 251.
4538
Epoch 148/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 9ms/step - loss: 43891.9492 - root_mean_squared_error: 209.5001 - val_loss: 64100.9141 - val_root_mean_squared_error: 253.1
783
```

```
[30]: {'model': 'Neural Network',
 'best_params': {'epochs': 300, 'batch_size': 64},
 'cv_rmse': None,
 'train_rmse': 178.1974728997621,
 'test_rmse': 221.42547859900695,
 'train_r2': 0.9423158078005699,
 'test_r2': 0.9061021519495637,
 'test_mae': 142.0474225058674,
 'overfit_gap_r2': 0.036213655059006266}
```

The Neural Network that we modeled performs better than most of the models, but still it doesn't achieve the results of XGBoost and RandomForest. During training we notice that it converges around 148 epochs, so we cannot improve it further unless we model a better NN.

Part 2: Trying models with partial data set

During EDA and other analysis we noticed that some of the features had unexpected effect or not a very clear effect, so I would say, it would be best to also try training some ML model without these features and see how they perform. We might find out that the model might be better of without these noisy features and it also makes it better for the users not having to add every single unnecessary detail about the apartment.

First we try it without the features that had an unexpected opposite effect.

```
[31]: features_reduced = [
    c for c in X_train.columns
    if c not in ["has_bathtub", "has_terrasse"]
]

X_train_2 = X_train[features_reduced]
X_test_2 = X_test[features_reduced]
```

```
[32]: features_2 = X_train_2.columns.tolist()

metrics_rf_2, bestmodel_rf_2 = evaluate_model(
    "Random Forest - no bathtub/terrasse",
    RandomForestRegressor(random_state=42),
    {
        "model__n_estimators": [100, 200, 300, 400],
        "model__max_depth": [None, 10, 20, 30, 40],
        "model__min_samples_split": [2, 5, 10, 20],
        "model__min_samples_leaf": [1, 2, 4, 8],
        "model__max_features": ["sqrt", "log2"],
    },
    X_train_2,
    y_train,
    X_test_2,
    y_test,
    feature_cols=features_2
)

metrics_rf_2
```

```
[32]: {'model': 'Random Forest - no bathtub/terrasse',
 'best_params': {'model__max_depth': 40,
  'model__max_features': 'sqrt',
  'model__min_samples_leaf': 1,
  'model__min_samples_split': 2,
  'model__n_estimators': 400},
 'cv_rmse': 234.2207994033461,
 'train_rmse': 95.45431000119082,
 'test_rmse': 206.311561869115,
 'train_r2': 0.9834482156697438,
 'test_r2': 0.9184831125194224,
 'test_mae': 123.62733364823701,
 'overfit_gap_r2': 0.06496510315032145}
```

Random forest seems to be performing slightly better so it seems that it would be better to not train our models without these features.

```
[33]: metrics_nn_2, bestmodel_nn_2, nn_pipeline_2, history_nn_2 = evaluate_nn(
    X_train_2,
    y_train,
    X_test_2,
    y_test,
    epochs=300,
    batch_size=64,
    feature_cols=features_2
)

metrics_nn_2
```

```
Epoch 1/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 3s 5ms/step - loss: 1827570.0000 - root_mean_squared_error: 1351.8762 - val_loss: 1484833.6250 - val_root_mean_squared_error:
1218.5374
Epoch 2/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 480540.7500 - root_mean_squared_error: 693.2100 - val_loss: 294701.6562 - val_root_mean_squared_error: 54
2.8636
Epoch 3/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 227515.6875 - root_mean_squared_error: 476.9853 - val_loss: 186626.1875 - val_root_mean_squared_error: 43
2.0017
Epoch 4/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 189834.6094 - root_mean_squared_error: 435.6993 - val_loss: 140226.3281 - val_root_mean_squared_error: 37
4.4671
Epoch 5/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 156056.7188 - root_mean_squared_error: 395.0392 - val_loss: 112875.5078 - val_root_mean_squared_error: 33
5.9683
Epoch 6/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 132211.4531 - root_mean_squared_error: 363.6079 - val_loss: 102048.2656 - val_root_mean_squared_error: 31
9.4487
Epoch 7/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 5ms/step - loss: 119072.7734 - root_mean_squared_error: 345.0681 - val_loss: 95470.6406 - val_root_mean_squared_error: 308.
9819
Epoch 8/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 5ms/step - loss: 111608.5312 - root_mean_squared_error: 334.0773 - val_loss: 90677.2656 - val_root_mean_squared_error: 301.
1253
Epoch 9/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 104773.0078 - root_mean_squared_error: 323.6853 - val_loss: 87355.9922 - val_root_mean_squared_error: 295.
5591
Epoch 10/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 6ms/step - loss: 100987.8047 - root_mean_squared_error: 317.7845 - val_loss: 83464.9531 - val_root_mean_squared_error: 288.
9016
Epoch 11/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 95954.4688 - root_mean_squared_error: 309.7638 - val_loss: 81738.4219 - val_root_mean_squared_error: 285.8
978
Epoch 12/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 91720.8594 - root_mean_squared_error: 302.8531 - val_loss: 84685.5781 - val_root_mean_squared_error: 291.0
064
Epoch 13/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 90068.0703 - root_mean_squared_error: 300.1121 - val_loss: 82519.0156 - val_root_mean_squared_error: 287.2
597
Epoch 14/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 89407.5469 - root_mean_squared_error: 299.0095 - val_loss: 80287.8672 - val_root_mean_squared_error: 283.3
496
Epoch 15/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 83032.7344 - root_mean_squared_error: 288.1525 - val_loss: 79211.8750 - val_root_mean_squared_error: 281.4
444
Epoch 16/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 84021.3516 - root_mean_squared_error: 289.8629 - val_loss: 76865.0547 - val_root_mean_squared_error: 277.2
439
Epoch 17/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 82088.6250 - root_mean_squared_error: 286.5096 - val_loss: 74341.1406 - val_root_mean_squared_error: 272.6
541
Epoch 18/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 79180.0156 - root_mean_squared_error: 281.3878 - val_loss: 75430.4219 - val_root_mean_squared_error: 274.6
443
Epoch 19/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 77165.3047 - root_mean_squared_error: 277.7848 - val_loss: 74043.0547 - val_root_mean_squared_error: 272.1
068
Epoch 20/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 75157.2188 - root_mean_squared_error: 274.1465 - val_loss: 75278.1562 - val_root_mean_squared_error: 274.3
669
Epoch 21/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 74908.2266 - root_mean_squared_error: 273.6920 - val_loss: 75346.6328 - val_root_mean_squared_error: 274.4
917
Epoch 22/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 72906.2109 - root_mean_squared_error: 270.0098 - val_loss: 73536.4297 - val_root_mean_squared_error: 271.1
743
Epoch 23/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 71016.1328 - root_mean_squared_error: 266.4868 - val_loss: 75775.3047 - val_root_mean_squared_error: 275.2
714
Epoch 24/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 69718.3203 - root_mean_squared_error: 264.0405 - val_loss: 72771.8594 - val_root_mean_squared_error: 269.7
608
Epoch 25/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 70201.9688 - root_mean_squared_error: 264.9547 - val_loss: 70692.5703 - val_root_mean_squared_error: 265.8
789
Epoch 26/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 71392.7500 - root_mean_squared_error: 267.1924 - val_loss: 73348.2109 - val_root_mean_squared_error: 270.8
270
Epoch 27/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 68864.4844 - root_mean_squared_error: 262.4186 - val_loss: 71087.1328 - val_root_mean_squared_error: 266.6
199
Epoch 28/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 68251.4844 - root_mean_squared_error: 261.2480 - val_loss: 73397.0859 - val_root_mean_squared_error: 270.9
172
Epoch 29/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 67210.0625 - root_mean_squared_error: 259.2472 - val_loss: 68927.3281 - val_root_mean_squared_error: 262.5
383
Epoch 30/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 66141.1172 - root_mean_squared_error: 257.1772 - val_loss: 72377.9609 - val_root_mean_squared_error: 269.0
297
Epoch 31/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 65788.1797 - root_mean_squared_error: 256.4902 - val_loss: 71313.3125 - val_root_mean_squared_error: 267.0
437
Epoch 32/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 64390.3047 - root_mean_squared_error: 253.7505 - val_loss: 68015.6641 - val_root_mean_squared_error: 260.7
962
Epoch 33/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 62241.5859 - root_mean_squared_error: 249.4806 - val_loss: 67687.6328 - val_root_mean_squared_error: 260.1
665
Epoch 34/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 62898.9961 - root_mean_squared_error: 250.7947 - val_loss: 71119.8281 - val_root_mean_squared_error: 266.6
811
Epoch 35/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 65520.5078 - root_mean_squared_error: 255.9678 - val_loss: 68874.0234 - val_root_mean_squared_error: 262.4
367
Epoch 36/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 61931.3047 - root_mean_squared_error: 248.8580 - val_loss: 69227.5781 - val_root_mean_squared_error: 263.1
094
Epoch 37/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 62444.9062 - root_mean_squared_error: 249.8877 - val_loss: 67673.9297 - val_root_mean_squared_error: 260.1
402
Epoch 38/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 64009.1289 - root_mean_squared_error: 252.9982 - val_loss: 69984.0391 - val_root_mean_squared_error: 264.5
430
Epoch 39/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 59681.4375 - root_mean_squared_error: 244.2957 - val_loss: 67302.0469 - val_root_mean_squared_error: 259.4
244
Epoch 40/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 61187.3828 - root_mean_squared_error: 247.3588 - val_loss: 68148.7734 - val_root_mean_squared_error: 261.0
512
Epoch 41/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 63565.2461 - root_mean_squared_error: 252.1194 - val_loss: 68691.8125 - val_root_mean_squared_error: 262.0
893
Epoch 42/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 59501.8047 - root_mean_squared_error: 243.9278 - val_loss: 70379.2969 - val_root_mean_squared_error: 265.2
890
Epoch 43/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 59560.2383 - root_mean_squared_error: 244.0475 - val_loss: 70020.6328 - val_root_mean_squared_error: 264.6
122
Epoch 44/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 59136.4805 - root_mean_squared_error: 243.1777 - val_loss: 67997.6172 - val_root_mean_squared_error: 260.7
615
Epoch 45/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 58257.0508 - root_mean_squared_error: 241.3628 - val_loss: 67107.5000 - val_root_mean_squared_error: 259.0
491
Epoch 46/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 58098.8594 - root_mean_squared_error: 241.0349 - val_loss: 70752.2188 - val_root_mean_squared_error: 265.9
909
Epoch 47/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 59489.1680 - root_mean_squared_error: 243.9018 - val_loss: 67903.7578 - val_root_mean_squared_error: 260.5
814
Epoch 48/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 57439.8828 - root_mean_squared_error: 239.6640 - val_loss: 65922.5938 - val_root_mean_squared_error: 256.7
519
Epoch 49/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 55911.8203 - root_mean_squared_error: 236.4545 - val_loss: 66388.7188 - val_root_mean_squared_error: 257.6
500
Epoch 50/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 57301.0430 - root_mean_squared_error: 239.3741 - val_loss: 65555.5703 - val_root_mean_squared_error: 256.0
361
Epoch 51/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 56805.9844 - root_mean_squared_error: 238.3378 - val_loss: 67067.8906 - val_root_mean_squared_error: 258.9
726
Epoch 52/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 5ms/step - loss: 54350.8633 - root_mean_squared_error: 233.1304 - val_loss: 67785.9922 - val_root_mean_squared_error: 260.3
553
Epoch 53/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 56189.4023 - root_mean_squared_error: 237.0407 - val_loss: 66209.9609 - val_root_mean_squared_error: 257.3
109
Epoch 54/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 56566.9492 - root_mean_squared_error: 237.8358 - val_loss: 67271.9453 - val_root_mean_squared_error: 259.3
662
Epoch 55/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 55303.9023 - root_mean_squared_error: 235.1655 - val_loss: 66283.0469 - val_root_mean_squared_error: 257.4
528
Epoch 56/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 54034.8555 - root_mean_squared_error: 232.4516 - val_loss: 65770.4922 - val_root_mean_squared_error: 256.4
554
Epoch 57/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 53816.4688 - root_mean_squared_error: 231.9814 - val_loss: 66111.4297 - val_root_mean_squared_error: 257.1
193
Epoch 58/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 54581.2461 - root_mean_squared_error: 233.6240 - val_loss: 65932.5000 - val_root_mean_squared_error: 256.7
711
Epoch 59/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 53268.3672 - root_mean_squared_error: 230.7970 - val_loss: 65886.6484 - val_root_mean_squared_error: 256.6
818
Epoch 60/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 56143.2930 - root_mean_squared_error: 236.9434 - val_loss: 65419.3750 - val_root_mean_squared_error: 255.7
699
Epoch 61/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 5ms/step - loss: 53066.8477 - root_mean_squared_error: 230.3600 - val_loss: 65061.6445 - val_root_mean_squared_error: 255.0
696
Epoch 62/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 8ms/step - loss: 52638.5195 - root_mean_squared_error: 229.4284 - val_loss: 63478.7539 - val_root_mean_squared_error: 251.9
477
Epoch 63/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 11ms/step - loss: 55522.3672 - root_mean_squared_error: 235.6295 - val_loss: 63730.1172 - val_root_mean_squared_error: 252.
4460
Epoch 64/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 10ms/step - loss: 52771.7188 - root_mean_squared_error: 229.7184 - val_loss: 66544.3750 - val_root_mean_squared_error: 257.
9597
Epoch 65/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 6ms/step - loss: 52960.7891 - root_mean_squared_error: 230.1296 - val_loss: 64910.3047 - val_root_mean_squared_error: 254.7
728
Epoch 66/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 53301.6172 - root_mean_squared_error: 230.8689 - val_loss: 65798.9062 - val_root_mean_squared_error: 256.5
107
Epoch 67/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 50630.1875 - root_mean_squared_error: 225.0090 - val_loss: 65937.5469 - val_root_mean_squared_error: 256.7
808
Epoch 68/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 52859.1484 - root_mean_squared_error: 229.9086 - val_loss: 65747.3984 - val_root_mean_squared_error: 256.4
102
Epoch 69/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 52178.3789 - root_mean_squared_error: 228.4234 - val_loss: 65162.8047 - val_root_mean_squared_error: 255.2
678
Epoch 70/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 50495.3203 - root_mean_squared_error: 224.7091 - val_loss: 66680.6172 - val_root_mean_squared_error: 258.2
236
Epoch 71/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 53583.5898 - root_mean_squared_error: 231.4788 - val_loss: 63515.1953 - val_root_mean_squared_error: 252.0
199
Epoch 72/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 51620.1602 - root_mean_squared_error: 227.1981 - val_loss: 61737.8750 - val_root_mean_squared_error: 248.4
687
Epoch 73/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 51000.2969 - root_mean_squared_error: 225.8298 - val_loss: 61039.4336 - val_root_mean_squared_error: 247.0
592
Epoch 74/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 50535.3828 - root_mean_squared_error: 224.7981 - val_loss: 63800.3203 - val_root_mean_squared_error: 252.5
849
Epoch 75/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 5ms/step - loss: 48138.6484 - root_mean_squared_error: 219.4025 - val_loss: 62144.2227 - val_root_mean_squared_error: 249.2
850
Epoch 76/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 50536.5117 - root_mean_squared_error: 224.8006 - val_loss: 61882.8672 - val_root_mean_squared_error: 248.7
603
Epoch 77/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 50004.8359 - root_mean_squared_error: 223.6149 - val_loss: 62830.3242 - val_root_mean_squared_error: 250.6
574
Epoch 78/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 7ms/step - loss: 48860.8359 - root_mean_squared_error: 221.0421 - val_loss: 63280.4219 - val_root_mean_squared_error: 251.5
536
Epoch 79/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 5ms/step - loss: 50042.1953 - root_mean_squared_error: 223.6984 - val_loss: 65096.0977 - val_root_mean_squared_error: 255.1
370
Epoch 80/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 50021.7148 - root_mean_squared_error: 223.6527 - val_loss: 62901.9336 - val_root_mean_squared_error: 250.8
002
Epoch 81/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 50168.0078 - root_mean_squared_error: 223.9794 - val_loss: 60598.5391 - val_root_mean_squared_error: 246.1
653
Epoch 82/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 50173.1758 - root_mean_squared_error: 223.9910 - val_loss: 61589.6016 - val_root_mean_squared_error: 248.1
701
Epoch 83/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 50377.0312 - root_mean_squared_error: 224.4456 - val_loss: 63925.4609 - val_root_mean_squared_error: 252.8
324
Epoch 84/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 49275.5352 - root_mean_squared_error: 221.9782 - val_loss: 61592.2969 - val_root_mean_squared_error: 248.1
755
Epoch 85/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 6ms/step - loss: 48687.4336 - root_mean_squared_error: 220.6495 - val_loss: 62474.0195 - val_root_mean_squared_error: 249.9
456
Epoch 86/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 49104.1641 - root_mean_squared_error: 221.5918 - val_loss: 66219.7109 - val_root_mean_squared_error: 257.3
295
Epoch 87/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 50010.5938 - root_mean_squared_error: 223.6276 - val_loss: 62029.8711 - val_root_mean_squared_error: 249.0
555
Epoch 88/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 49100.1289 - root_mean_squared_error: 221.5827 - val_loss: 64135.0156 - val_root_mean_squared_error: 253.2
070
Epoch 89/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 47838.5859 - root_mean_squared_error: 218.7175 - val_loss: 62344.5312 - val_root_mean_squared_error: 249.6
864
Epoch 90/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 47520.2070 - root_mean_squared_error: 217.9884 - val_loss: 65429.8008 - val_root_mean_squared_error: 255.7
900
Epoch 91/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 5ms/step - loss: 48358.3945 - root_mean_squared_error: 219.9026 - val_loss: 63050.9062 - val_root_mean_squared_error: 251.0
969
Epoch 92/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 6ms/step - loss: 46948.4023 - root_mean_squared_error: 216.6729 - val_loss: 62584.7578 - val_root_mean_squared_error: 250.1
669
Epoch 93/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 7ms/step - loss: 47590.3516 - root_mean_squared_error: 218.1492 - val_loss: 61518.9336 - val_root_mean_squared_error: 248.0
276
Epoch 94/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 2s 8ms/step - loss: 47357.4102 - root_mean_squared_error: 217.6147 - val_loss: 60256.1914 - val_root_mean_squared_error: 245.4
688
Epoch 95/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - loss: 47826.2461 - root_mean_squared_error: 218.6891 - val_loss: 60102.7617 - val_root_mean_squared_error: 245.1
561
Epoch 96/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 7ms/step - loss: 47172.7930 - root_mean_squared_error: 217.1900 - val_loss: 60250.5547 - val_root_mean_squared_error: 245.4
573
Epoch 97/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 5ms/step - loss: 47550.0664 - root_mean_squared_error: 218.0569 - val_loss: 60075.4883 - val_root_mean_squared_error: 245.1
004
Epoch 98/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 6ms/step - loss: 48210.4531 - root_mean_squared_error: 219.5658 - val_loss: 59286.1914 - val_root_mean_squared_error: 243.4
849
Epoch 99/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 6ms/step - loss: 47803.2305 - root_mean_squared_error: 218.6365 - val_loss: 62602.8359 - val_root_mean_squared_error: 250.2
030
Epoch 100/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 47097.6211 - root_mean_squared_error: 217.0169 - val_loss: 59987.6445 - val_root_mean_squared_error: 244.9
211
Epoch 101/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 46692.2031 - root_mean_squared_error: 216.0808 - val_loss: 62639.8125 - val_root_mean_squared_error: 250.2
769
Epoch 102/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 47334.1016 - root_mean_squared_error: 217.5610 - val_loss: 61555.4883 - val_root_mean_squared_error: 248.1
012
Epoch 103/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 46149.6055 - root_mean_squared_error: 214.8216 - val_loss: 60015.5234 - val_root_mean_squared_error: 244.9
780
Epoch 104/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 45788.3281 - root_mean_squared_error: 213.9790 - val_loss: 61070.7500 - val_root_mean_squared_error: 247.1
223
Epoch 105/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 5ms/step - loss: 46459.8672 - root_mean_squared_error: 215.5425 - val_loss: 61507.6680 - val_root_mean_squared_error: 248.0
048
Epoch 106/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 5ms/step - loss: 45914.2734 - root_mean_squared_error: 214.2731 - val_loss: 61821.3125 - val_root_mean_squared_error: 248.6
363
Epoch 107/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 46002.1328 - root_mean_squared_error: 214.4780 - val_loss: 59718.6133 - val_root_mean_squared_error: 244.3
712
Epoch 108/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 46757.5898 - root_mean_squared_error: 216.2320 - val_loss: 58918.6094 - val_root_mean_squared_error: 242.7
289
Epoch 109/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 45744.1211 - root_mean_squared_error: 213.8756 - val_loss: 61351.8477 - val_root_mean_squared_error: 247.6
904
Epoch 110/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 46005.3086 - root_mean_squared_error: 214.4854 - val_loss: 57438.9609 - val_root_mean_squared_error: 239.6
615
Epoch 111/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 45517.3008 - root_mean_squared_error: 213.3447 - val_loss: 60576.8516 - val_root_mean_squared_error: 246.1
210
Epoch 112/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 45868.1992 - root_mean_squared_error: 214.1655 - val_loss: 62025.5352 - val_root_mean_squared_error: 249.0
466
Epoch 113/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 46268.4922 - root_mean_squared_error: 215.0980 - val_loss: 62260.5039 - val_root_mean_squared_error: 249.5
179
Epoch 114/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 45345.1016 - root_mean_squared_error: 212.9408 - val_loss: 62499.4531 - val_root_mean_squared_error: 249.9
962
Epoch 115/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 44439.1289 - root_mean_squared_error: 210.8027 - val_loss: 59670.0430 - val_root_mean_squared_error: 244.2
718
Epoch 116/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 44753.8555 - root_mean_squared_error: 211.5479 - val_loss: 60119.4023 - val_root_mean_squared_error: 245.1
898
Epoch 117/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 45682.7891 - root_mean_squared_error: 213.7321 - val_loss: 60494.9492 - val_root_mean_squared_error: 245.9
545
Epoch 118/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 44362.6055 - root_mean_squared_error: 210.6211 - val_loss: 59244.8164 - val_root_mean_squared_error: 243.3
998
Epoch 119/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 45732.9062 - root_mean_squared_error: 213.8494 - val_loss: 61267.8359 - val_root_mean_squared_error: 247.5
207
Epoch 120/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 46473.6953 - root_mean_squared_error: 215.5744 - val_loss: 59936.1875 - val_root_mean_squared_error: 244.8
159
Epoch 121/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 44588.7734 - root_mean_squared_error: 211.1573 - val_loss: 60043.9141 - val_root_mean_squared_error: 245.0
358
Epoch 122/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 44886.9688 - root_mean_squared_error: 211.6733 - val_loss: 62084.6562 - val_root_mean_squared_error: 249.1
652
Epoch 123/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 44474.0039 - root_mean_squared_error: 210.8854 - val_loss: 60212.9141 - val_root_mean_squared_error: 245.3
804
Epoch 124/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 42580.9648 - root_mean_squared_error: 206.3482 - val_loss: 62011.1758 - val_root_mean_squared_error: 249.0
177
Epoch 125/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 43634.5469 - root_mean_squared_error: 208.8856 - val_loss: 60951.0312 - val_root_mean_squared_error: 246.8
799
Epoch 126/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 44805.6367 - root_mean_squared_error: 211.6702 - val_loss: 58463.8672 - val_root_mean_squared_error: 241.7
902
Epoch 127/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 5ms/step - loss: 44438.5195 - root_mean_squared_error: 210.8012 - val_loss: 61012.6055 - val_root_mean_squared_error: 247.0
045
Epoch 128/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 43956.9453 - root_mean_squared_error: 209.6558 - val_loss: 59931.8945 - val_root_mean_squared_error: 244.8
071
Epoch 129/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 44854.4336 - root_mean_squared_error: 211.7854 - val_loss: 59526.8906 - val_root_mean_squared_error: 243.9
785
Epoch 130/300
156/156 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - loss: 44778.5898 - root_mean_squared_error: 211.6062 - val_loss: 66757.0781 - val_root_mean_squared_error: 258.3
712
```

```
[33]: {'model': 'Neural Network',
 'best_params': {'epochs': 300, 'batch_size': 64},
 'cv_rmse': None,
 'train_rmse': 180.64100974059124,
 'test_rmse': 222.5471662644276,
 'train_r2': 0.9407229697047931,
 'test_r2': 0.905148414993896,
 'test_mae': 144.113650951569,
 'overfit_gap_r2': 0.0355745547108971}
```

We cannot reach the same conclusion about the neural network, which performs sligtly worse without these 2.

Now we try it by also removing the other noisy features.

```
[10]: features_reduced = [
    c for c in X_train.columns
    if c not in ["has_bathtub", "has_terrasse", "has_balkon", "has_garten", "parking", "has_basement", "is_barrier-free"]
]

X_train_3 = X_train[features_reduced]
X_test_3 = X_test[features_reduced]

features_3 = X_train_3.columns.tolist()
```

```
[35]: metrics_rf_3, bestmodel_rf_3 = evaluate_model(
    "Random Forest - without 7 least important features.",
    RandomForestRegressor(random_state=42),
    {
        "model__n_estimators": [100, 200, 300, 400],
        "model__max_depth": [None, 10, 20, 30, 40],
        "model__min_samples_split": [2, 5, 10, 20],
        "model__min_samples_leaf": [1, 2, 4, 8],
        "model__max_features": ["sqrt", "log2"],
    },
    X_train_3,
    y_train,
    X_test_3,
    y_test,
    feature_cols=features_3
)

metrics_rf_3
```

```
[35]: {'model': 'Random Forest - without 7 least important features.',
 'best_params': {'model__max_depth': 40,
  'model__max_features': 'sqrt',
  'model__min_samples_leaf': 1,
  'model__min_samples_split': 2,
  'model__n_estimators': 400},
 'cv_rmse': 233.66988212572176,
 'train_rmse': 95.08428516250468,
 'test_rmse': 204.44780700829875,
 'train_r2': 0.9835762916151471,
 'test_r2': 0.9199492568392196,
 'test_mae': 124.71615341677166,
 'overfit_gap_r2': 0.06362703477592757}
```

Random forest seems to be performing even better.

```
[36]: param_grid_xgb = {
    "model__n_estimators": [400, 600],
    "model__max_depth": [5, 7],
    "model__learning_rate": [0.01, 0.1],
    "model__subsample": [0.8, 1.0],
    "model__colsample_bytree": [0.8, 1.0],
    "model__min_child_weight": [1, 3],
    "model__gamma": [0, 0.1]
}

metrics_xgb_3, bestmodel_xgb_3 = evaluate_model(
    "XGBoost - without 7 least important features.",
    XGBRegressor(
        random_state=42,
        objective="reg:squarederror"
    ),
    param_grid_xgb,
    X_train_3,
    y_train,
    X_test_3,
    y_test,
    feature_cols=features_3
)

metrics_xgb_3
```

```
[36]: {'model': 'XGBoost - without 7 least important features.',
 'best_params': {'model__colsample_bytree': 1.0,
  'model__gamma': 0.1,
  'model__learning_rate': 0.1,
  'model__max_depth': 7,
  'model__min_child_weight': 1,
  'model__n_estimators': 600,
  'model__subsample': 0.8},
 'cv_rmse': 218.82289186195982,
 'train_rmse': 91.25260253268166,
 'test_rmse': 177.2901258608504,
 'train_r2': 0.9848732979614432,
 'test_r2': 0.9398037341128603,
 'test_mae': 107.31663482900917,
 'overfit_gap_r2': 0.04506956384858285}
```

And XGBoost performs much more better.

Part 3: Model comparison and picking the most suitable model

Now that we trained all the models, it would be better if we compare them in details before picking the final model, even though we noticed XGBoost performed the best.

```
[37]: results = [
    metrics_linear,
    metrics_rf,
    metrics_rf_2,
    metrics_xgb,
    metrics_knn,
    metrics_svr,
    metrics_nn,
    metrics_nn_2,
    metrics_rf_3,
    metrics_xgb_3
]

results_df = pd.DataFrame(results)

results_df
```

[37]:

	model	best_params	cv_rmse	train_rmse	test_rmse	train_r2	test_r2	test_mae	overfit_gap_r2
0	linear_regression	{}	432.648392	428.587877	408.987703	0.666317	0.679653	289.313141	-0.013335
1	Random Forest	{'model__max_depth': 30, 'model__max_features'...	234.580624	108.987300	209.193880	0.978422	0.916190	125.334425	0.062233
2	Random Forest - no bathtub/terrasse	{'model__max_depth': 40, 'model__max_features'...	234.220799	95.454310	206.311562	0.983448	0.918483	123.627334	0.064965
3	XGBoost	{'model__colsample_bytree': 0.8, 'model__gamma...	213.525741	115.660841	184.139173	0.975699	0.935063	111.740268	0.040636
4	knn	{'model__n_neighbors': 3, 'model__weights': 'd...	295.784776	46.105525	265.524793	0.996138	0.864976	153.741122	0.131162
5	SVR	{'model__regressor__C': 0.1, 'model__regressor...	437.011182	434.312876	408.331666	0.657343	0.680680	286.559764	-0.023336
6	Neural Network	{'epochs': 300, 'batch_size': 64}	NaN	178.197473	221.425479	0.942316	0.906102	142.047423	0.036214
7	Neural Network	{'epochs': 300, 'batch_size': 64}	NaN	180.641010	222.547166	0.940723	0.905148	144.113651	0.035575
8	Random Forest - without 7 least important feat...	{'model__max_depth': 40, 'model__max_features'...	233.669882	95.084285	204.447807	0.983576	0.919949	124.716153	0.063627
9	XGBoost - without 7 least important features.	{'model__colsample_bytree': 1.0, 'model__gamma...	218.822892	91.252603	177.290126	0.984873	0.939804	107.316635	0.045070

```
[38]: results_df = pd.DataFrame(results)[
    [
        "model",
        "cv_rmse",
        "train_rmse",
        "test_rmse",
        "train_r2",
        "test_r2",
        "test_mae",
        "overfit_gap_r2",
    ]
]

results_df
```

[38]:

	model	cv_rmse	train_rmse	test_rmse	train_r2	test_r2	test_mae	overfit_gap_r2
0	linear_regression	432.648392	428.587877	408.987703	0.666317	0.679653	289.313141	-0.013335
1	Random Forest	234.580624	108.987300	209.193880	0.978422	0.916190	125.334425	0.062233
2	Random Forest - no bathtub/terrasse	234.220799	95.454310	206.311562	0.983448	0.918483	123.627334	0.064965
3	XGBoost	213.525741	115.660841	184.139173	0.975699	0.935063	111.740268	0.040636
4	knn	295.784776	46.105525	265.524793	0.996138	0.864976	153.741122	0.131162
5	SVR	437.011182	434.312876	408.331666	0.657343	0.680680	286.559764	-0.023336
6	Neural Network	NaN	178.197473	221.425479	0.942316	0.906102	142.047423	0.036214
7	Neural Network	NaN	180.641010	222.547166	0.940723	0.905148	144.113651	0.035575
8	Random Forest - without 7 least important feat...	233.669882	95.084285	204.447807	0.983576	0.919949	124.716153	0.063627
9	XGBoost - without 7 least important features.	218.822892	91.252603	177.290126	0.984873	0.939804	107.316635	0.045070

```
[39]: results_df.loc[7, "model"] = "Neural Network - no bathtub/terrasse"
```

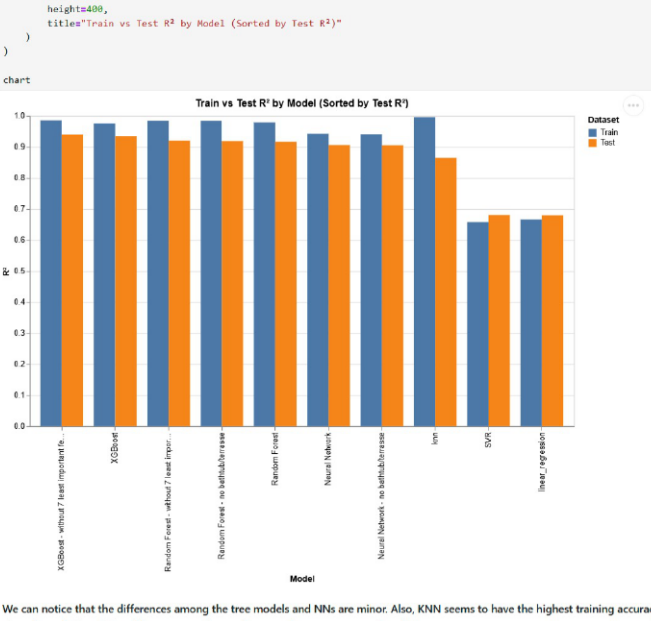
```
[40]: results_df = results_df.sort_values(
    "test_r2",
    ascending=False
)

results_df
```

[40]:

	model	cv_rmse	train_rmse	test_rmse	train_r2	test_r2	test_mae	overfit_gap_r2
9	XGBoost - without 7 least important features.	218.822892	91.252603	177.290126	0.984873	0.939804	107.316635	0.045070
3	XGBoost	213.525741	115.660841	184.139173	0.975699	0.935063	111.740268	0.040636
8	Random Forest - without 7 least important feat...	233.669882	95.084285	204.447807	0.983576	0.919949	124.716153	0.063627
2	Random Forest - no bathtub/terrasse	234.220799	95.454310	206.311562	0.983448	0.918483	123.627334	0.064965
1	Random Forest	234.580624	108.987300	209.193880	0.978422	0.916190	125.334425	0.062233
6	Neural Network	NaN	178.197473	221.425479	0.942316	0.906102	142.047423	0.036214
7	Neural Network - no bathtub/terrasse	NaN	180.641010	222.547166	0.940723	0.905148	144.113651	0.035575
4	knn	295.784776	46.105525	265.524793	0.996138	0.864976	153.741122	0.131162
5	SVR	437.011182	434.312876	408.331666	0.657343	0.680680	286.559764	-0.023336
0	linear_regression	432.648392	428.587877	408.987703	0.666317	0.679653	289.313141	-0.013335

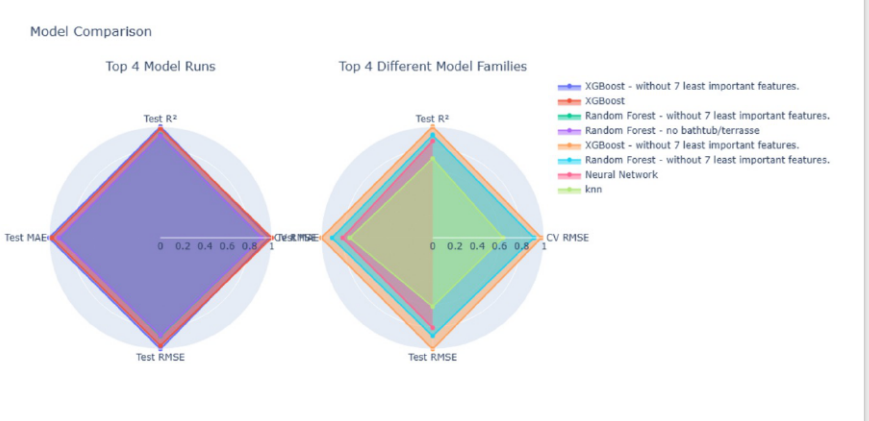
```
[41]: # ordered by test_r2 descending
model_order = results_df.sort_values(by="test_r2", ascending=False)["model"].tolist()

# reshape data to long format
results_long = results_df.melt(
    id_vars="model",
    value_vars=["train_r2", "test_r2"],
    var_name="dataset",
    value_name="r2"
)

results_long["dataset"] = results_long["dataset"].replace({
    "train_r2": "Train",
    "test_r2": "Test"
})

dataset_order = ["Train", "Test"]

chart = (
    alt.Chart(results_long)
    .mark_bar()
    .encode(
        x=alt.X(
            "model:N",
            title="Model",
            sort=model_order  # Explicitly set x-axis sort order by test_r2
        ),
        xOffset=alt.XOffset(
            "dataset:N",
            title="Dataset",
            sort=dataset_order
        ),
        y=alt.Y(
            "r2:Q",
            title="R²",
            scale=alt.Scale(domain=[0, 1])
        ),
        color=alt.Color(
            "dataset:N",
            title="Dataset",
            sort=dataset_order
        ),
        tooltip=[
            alt.Tooltip("model:N", title="Model"),
            alt.Tooltip("dataset:N", title="Dataset"),
            alt.Tooltip("r2:Q", title="R²", format=".3f")
        ]
    )
    .properties(
        width=700,
        height=400,
        title="Train vs Test R² by Model (Sorted by Test R²)"
    )
)

chart
```

[41]:



We can notice that the differences among the tree models and NNs are minor. Also, KNN seems to have the highest training accuracy out of all of them, that is clearly a sign of overfitting. SVR and linear regression perform much worse compared to the rest.

```
[43]: import plotly.graph_objects as go
from plotly.subplots import make_subplots

def _model_family(model_name):
    return model_name.split(" - ", 1)[0]


def build_radar(selected_results, all_results, title):
    metrics = ["test_r2", "cv_rmse", "test_rmse", "test_mae"]
    labels = ["Test R²", "CV RMSE", "Test RMSE", "Test MAE"]
    higher_is_better = [True, False, False, False]

    # Make a normalized copy of all results
    normalized = all_results.copy()

    # Normalize each metric
    for metric, higher in zip(metrics, higher_is_better):

        lo = normalized[metric].min()
        hi = normalized[metric].max()

        normalized[f"_radar_{metric}"] = (
            1.0 if hi == lo else
            (normalized[metric] - lo) / (hi - lo)
        )

        # Invert metrics where lower is better
        if not higher:
            normalized[f"_radar_{metric}"] = (
                1 - normalized[f"_radar_{metric}"]
            )

    # Get the normalized versions of the selected models
    selected_normalized = normalized.loc[selected_results.index]

    fig = go.Figure()

    for _, row in selected_normalized.iterrows():

        values = [
            float(row[f"_radar_{metric}"])
            for metric in metrics
        ]

        values.append(values[0])

        fig.add_trace(go.Scatterpolar(
            r=values,
            theta=labels + [labels[0]],
            fill="toself",
            name=row["model"],
        ))

    fig.update_layout(
        title=title,
        polar=dict(
            radialaxis=dict(
                visible=True,
                range=[0, 1]
            )
        ),
        showlegend=True,
    )

    return fig


# Keep only the best run from each model family.
best_by_family = (
    results_df.assign(family=results_df["model"].map(_model_family))
    .sort_values("test_r2", ascending=False)
    .drop_duplicates("family")
    .drop(columns="family")
)

top_4_models = results_df.head(4).copy()
fig_top4 = build_radar(top_4_models, results_df, "Top 4 Model Runs")

top_4_families = best_by_family.head(4).copy()
fig_families4 = build_radar(
    top_4_families,
    results_df,
    "Top 4 Different Model Families",
)

combined_fig = make_subplots(
    rows=1,
    cols=2,
    specs=[[{"type": "polar"}, {"type": "polar"}]],
    subplot_titles=["Top 4 Model Runs", "Top 4 Different Model Families"],
)

for trace in fig_top4.data:
    combined_fig.add_trace(trace, row=1, col=1)

for trace in fig_families4.data:
    combined_fig.add_trace(trace, row=1, col=2)

combined_fig.update_layout(
    width=1200,
    height=600,
    title="Model Comparison",
    showlegend=True,
)

combined_fig.update_polars(
    radialaxis=dict(
        visible=True,
        range=[0, 1],
        tickvals=[0, 0.2, 0.4, 0.6, 0.8, 1.0],
    ),
    angularaxis=dict(direction="clockwise"),
)

combined_fig.show()
```



It is good to consider all types metrics, not just R^2 for the testing set. Radar charts are a good type of visualization to display and compare model performance for different kind of metrics. In the first radar chart we compare the 4 best performing models, 2 XGBoosts and 2 RandomForests. XGBoost without the 7 features seems to perform better in all 4 metrics.
To get a view of how different model families perform, in the 2nd graph, we compare XGBoost, RandomForest, Neural Network and knn. Again, XGBoost permorms the best in all metrics except in Generalization Gap. Neural Network seems to have the smallest Genralization Gap. This could suggest that the Neural Network model could deal better with overfitting.
Also it is important to note that the metrics here are normalized and their directions are adjusted to show that the higher the value from 0 to 1, the better the metric is. The original values are still displayed on hover.

One very imporant thing about our project is that we don't just want to estimate a single point, but rather also suggest the user the interval of prices where their true could lie. For that we need some kind of uncertainty or confidence from the model and the standart XGBoost doesn't provide that, for this reason we need to train a variant of XGBoost called 'XGBoost Quantile Regression'. There is always a trade-off between things and therefore this model might cost us some prediction accuracy.

```
[44]: from sklearn.pipeline import Pipeline
from xgboost import XGBRegressor

quantile_pipeline = Pipeline([
    ("preprocessor", build_preprocessor(features_2)),
    ("model", XGBRegressor(
        objective="reg:quantileerror",
        quantile_alpha=0.5,
        n_estimators=600,
        max_depth=7,
        learning_rate=0.05,
        subsample=0.8,
        colsample_bytree=0.8,
        random_state=42,
    ))
])

quantile_pipeline.fit(X_train_2, y_train)

y_pred_q50 = quantile_pipeline.predict(X_test_2)

y_pred_q50 = np.asarray(y_pred_q50).ravel()
```

```
[45]: from sklearn.metrics import r2_score, mean_squared_error, mean_absolute_error

r2_q50 = r2_score(y_test, y_pred_q50)
rmse_q50 = np.sqrt(mean_squared_error(y_test, y_pred_q50))
mae_q50 = mean_absolute_error(y_test, y_pred_q50)

print("Quantile XGBoost (50%)")
print("Test R²:", r2_q50)
print("Test RMSE:", rmse_q50)
print("Test MAE:", mae_q50)
```

```
Quantile XGBoost (50%)
Test R²: 0.8869729524893049
Test RMSE: 242.9354895972518
Test MAE: 144.59292423700967
```

It seems like the model seems to be performing much poorer than the original XGBoost and even poorer than RandomForest. There is no point in keeping XGBoost, when RandomForest itself natively provides a way to estimate some model uncertainity and intervals of predicted values, therefore we will choose Random Forest as our final model.

Let's try to tune and select an optimal number of estimators and deepth without overfitting the model.

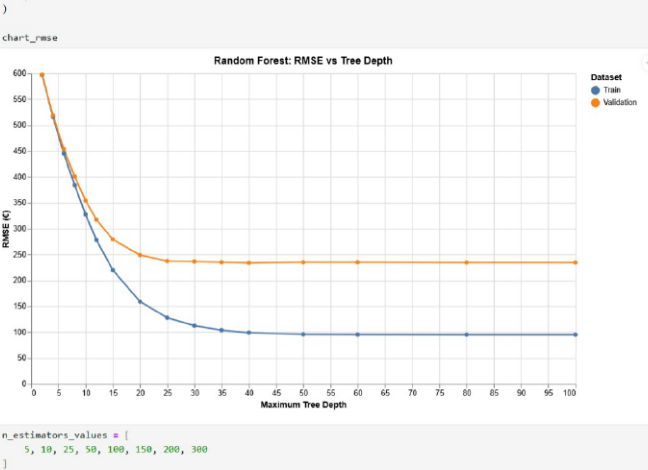
```
[13]: from sklearn.model_selection import cross_validate
from sklearn.metrics import make_scorer, mean_squared_error

depths = [2, 4, 6, 8, 10, 12, 15, 20, 25, 30, 35, 40, 50, 60, 80, 100]

depth_results = []

for depth in depths:
    rf_pipeline = build_pipeline(
        RandomForestRegressor(
            n_estimators=200,
            max_depth=depth,
            min_samples_split=2,
            min_samples_leaf=1,
            max_features="sqrt",
            random_state=42,
            n_jobs=-1
        ),
        feature_cols=features_3
    )

    scores = cross_validate(
        rf_pipeline,
        X_train_3,
        y_train,
        cv=make_cv(),
        scoring={
            "rmse": "neg_root_mean_squared_error",
            "r2": "r2"
        },
        return_train_score=True,
        n_jobs=-1
    )

    depth_results.append({
        "max_depth": depth,

        "train_rmse": -scores["train_rmse"].mean(),
        "val_rmse": -scores["test_rmse"].mean(),

        "train_r2": scores["train_r2"].mean(),
        "val_r2": scores["test_r2"].mean(),

        "rmse_gap": (
            -scores["test_rmse"].mean()
            - (-scores["train_rmse"].mean())
        ),

        "r2_gap": (
            scores["train_r2"].mean()
            - scores["test_r2"].mean()
        )
    })

depth_results = pd.DataFrame(depth_results)

depth_results
```

[13]:

	max_depth	train_rmse	val_rmse	train_r2	val_r2	rmse_gap	r2_gap
0	2	596.982331	597.551615	0.352565	0.351087	0.569284	0.001478
1	4	515.911958	518.863284	0.516469	0.510661	2.951325	0.005807
2	6	445.278032	453.944903	0.639757	0.625411	8.666871	0.014346
3	8	384.092772	400.927604	0.731970	0.707683	16.834832	0.024287
4	10	327.420251	354.174164	0.805227	0.771778	26.753913	0.033449
5	12	278.064077	317.210150	0.859533	0.816807	39.146073	0.042726
6	15	220.078897	279.346229	0.912003	0.857893	59.267332	0.054110
7	20	158.910927	248.932322	0.954120	0.887194	90.021395	0.066926
8	25	127.920046	237.471056	0.970272	0.897347	109.551010	0.072925
9	30	112.459079	236.584145	0.977021	0.898106	124.125066	0.078915
10	35	103.888801	235.154662	0.980389	0.899364	131.265860	0.081025
11	40	99.137719	234.018353	0.982142	0.900314	134.880634	0.081828
12	50	95.803233	235.124231	0.983322	0.899366	139.320998	0.083956
13	60	95.347683	235.192808	0.983480	0.899325	139.845125	0.084156
14	80	95.202363	234.680365	0.983531	0.899775	139.478002	0.083756
15	100	95.202363	234.680365	0.983531	0.899775	139.478002	0.083756

```
[14]: depth_long = depth_results.melt(
    id_vars="max_depth",
    value_vars=["train_rmse", "val_rmse"],
    var_name="dataset",
    value_name="rmse"
)

depth_long["dataset"] = depth_long["dataset"].replace({
    "train_rmse": "Train",
    "val_rmse": "Validation"
})

chart_rmse = (
    alt.Chart(depth_long)
    .mark_line(point=True)
    .encode(
        x=alt.X(
            "max_depth:Q",
            title="Maximum Tree Depth"
        ),
        y=alt.Y(
            "rmse:Q",
            title="RMSE (€)"
        ),
        color=alt.Color(
            "dataset:N",
            title="Dataset"
        ),
        tooltip=[
            alt.Tooltip("max_depth:Q", title="Max Depth"),
            alt.Tooltip("dataset:N", title="Dataset"),
            alt.Tooltip("rmse:Q", title="RMSE", format=".2f")
        ]
    )
    .properties(
        width=700,
        height=400,
        title="Random Forest: RMSE vs Tree Depth"
    )
)

chart_rmse
```

[14]:



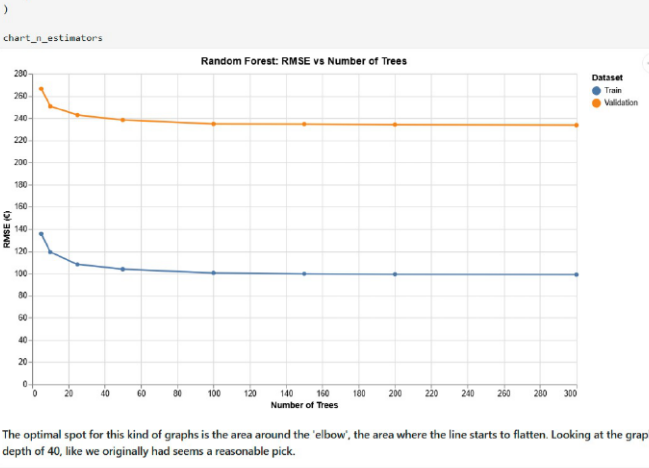
```
[16]: n_estimators_values = [
    5, 10, 25, 50, 100, 150, 200, 300
]

n_estimators_results = []

for n in n_estimators_values:

    rf_pipeline = build_pipeline(
        RandomForestRegressor(
            n_estimators=n,
            max_depth=40,           # fixed
            min_samples_split=2,    # fixed
            min_samples_leaf=1,     # fixed
            max_features="sqrt",    # fixed
            random_state=42,
            n_jobs=-1
        ),
        feature_cols=features_3
    )

    scores = cross_validate(
        rf_pipeline,
        X_train_3,
        y_train,
        cv=make_cv(),
        scoring={
            "rmse": "neg_root_mean_squared_error",
            "r2": "r2"
        },
        return_train_score=True,
        n_jobs=-1
    )

    n_estimators_results.append({
        "n_estimators": n,
        "train_rmse": -scores["train_rmse"].mean(),
        "val_rmse": -scores["test_rmse"].mean(),
        "train_r2": scores["train_r2"].mean(),
        "val_r2": scores["test_r2"].mean(),
        "rmse_gap": (
            -scores["test_rmse"].mean()
            - (-scores["train_rmse"].mean())
        ),
        "r2_gap": (
            scores["train_r2"].mean()
            - scores["test_r2"].mean()
        )
    })

n_estimators_results = pd.DataFrame(n_estimators_results)

n_estimators_results
```

[16]:

	n_estimators	train_rmse	val_rmse	train_r2	val_r2	rmse_gap	r2_gap
0	5	135.586990	266.437063	0.966599	0.870462	130.850074	0.096137
1	10	119.066764	250.651603	0.974237	0.885636	131.584838	0.088601
2	25	108.042347	242.743511	0.978785	0.892788	134.701164	0.085998
3	50	103.715908	238.272162	0.980454	0.896641	134.556254	0.083813
4	100	100.368777	234.674182	0.981695	0.899771	134.305406	0.081924
5	150	99.543143	234.523436	0.981995	0.899893	134.980294	0.082102
6	200	99.137719	234.018353	0.982142	0.900314	134.880634	0.081828
7	300	98.829187	233.604698	0.982252	0.900669	134.775511	0.081583

```
[17]: n_estimators_long = n_estimators_results.melt(
    id_vars="n_estimators",
    value_vars=["train_rmse", "val_rmse"],
    var_name="dataset",
    value_name="rmse"
)

n_estimators_long["dataset"] = n_estimators_long["dataset"].replace({
    "train_rmse": "Train",
    "val_rmse": "Validation"
})

chart_n_estimators = (
    alt.Chart(n_estimators_long)
    .mark_line(point=True)
    .encode(
        x=alt.X(
            "n_estimators:Q",
            title="Number of Trees"
        ),
        y=alt.Y(
            "rmse:Q",
            title="RMSE (€)"
        ),
        color=alt.Color(
            "dataset:N",
            title="Dataset"
        ),
        tooltip=[
            alt.Tooltip(
                "n_estimators:Q",
                title="Number of Trees"
            ),
            alt.Tooltip(
                "dataset:N",
                title="Dataset"
            ),
            alt.Tooltip(
                "rmse:Q",
                title="RMSE",
                format=".2f"
            )
        ]
    )
    .properties(
        width=700,
        height=400,
        title="Random Forest: RMSE vs Number of Trees"
    )
)

chart_n_estimators
```

[17]:



The optimal spot for this kind of graphs is the area around the 'elbow', the area where the line starts to flatten. Looking at the graphs, a number of estimators of 200 and a depth of 40, like we originally had seems a reasonable pick.

```
[18]: metrics_final, final_model = evaluate_model(
    "Random Forest - Final",
    RandomForestRegressor(
        n_estimators=200,
        max_depth=40,
        min_samples_split=2,
        min_samples_leaf=1,
        max_features="sqrt",
        random_state=42,
        n_jobs=-1
    ),
    {},
    X_train_3,
    y_train,
    X_test_3,
    y_test,
    feature_cols=features_3
)

metrics_final
```

```
[18]: {'model': 'Random Forest - Final',
 'best_params': {},
 'cv_rmse': 234.1715643410689,
 'train_rmse': 95.6914467030413,
 'test_rmse': 204.59704368790383,
 'train_r2': 0.9833658744738436,
 'test_r2': 0.9198323481038252,
 'test_mae': 124.74674650600103,
 'overfit_gap_r2': 0.06353352637001841}
```

Part 4: Saving the model

```
[19]: save_bundle(
    final_model,
    "final_model.pkl"
)
```

```
[19]: WindowsPath('final_model.pkl')
```